

MCKINLEY NATL PARK, AK, USA

WMO: 702645

Lat: 63.733N

Lon: 148.917W

Elev: 1720

StdP: 13.81

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 46403

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.6	-35.5	-43.7	0.5	-36.8	-35.8	0.8	-29.5	18.0	13.0	16.2	13.2	0.1	220	0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	74.7	56.7	71.8	55.7	67.8	54.3	58.9	70.6	57.0	67.6	55.2	65.1	6.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.2	66.6	61.4	52.3	62.0	59.7	50.4	57.8	58.4	26.3	70.8	25.0	68.0	24.0	64.7	65.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 15.4	12.9	11.8	DB	-44.5	80.3	5.3	3.9	-48.3	83.1	-51.4	85.4	-54.4	87.5	-58.3	90.4	
(5)			WB	-35.3	61.2	7.8	2.6	-40.9	63.0	-45.5	64.5	-49.9	65.9	-55.5	67.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	0.3	9.9	11.8	29.0	43.2	53.2	55.8	51.6	42.0	27.4	9.4	3.8	(6)
(7)		DBStd	23.74	22.24	17.18	14.47	10.40	7.00	5.08	4.66	5.67	7.35	11.57	16.66	19.14	(7)
(8)		HDD50	8387	1541	1124	1184	631	228	23	8	47	248	702	1218	1433	(8)
(9)		HDD65	13434	2006	1544	1649	1081	676	355	286	417	689	1165	1668	1898	(9)
(10)		CDD50	431	0	0	0	0	18	118	188	96	9	2	0	0	(10)
(11)		CDD65	2	0	0	0	0	0	1	1	0	0	0	0	0	(11)
(12)		CDH74	130	0	0	0	0	3	39	58	30	0	0	0	0	(12)
(13)		CDH80	10	0	0	0	0	0	3	5	2	0	0	0	0	(13)
(14)	Wind	WSAvg	4.1	3.3	3.9	4.9	5.1	5.0	4.2	3.9	3.7	4.0	4.3	3.4	3.3	(14)
(15)	Precipitation	PrecAvg	16.7	0.9	0.7	0.5	0.6	0.8	2.0	2.9	3.1	2.2	1.1	1.0	0.9	(15)
(16)		PrecMax	26.5	2.8	2.1	1.9	2.6	1.9	4.0	6.7	5.6	5.6	2.6	2.4	4.0	(16)
(17)		PrecMin	12.9	0.2	0.2	0.0	0.1	0.2	0.7	1.0	1.3	0.7	0.4	0.2	0.2	(17)
(18)		PrecStd	3.2	0.7	0.5	0.4	0.6	0.5	0.9	1.3	1.2	1.3	0.6	0.7	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	45.3	45.9	60.7	73.0	79.2	79.8	77.5	63.6	57.2	45.7	44.6	(19)
MCWB			38.6	35.5	36.3	44.9	52.4	57.0	59.0	58.7	50.0	45.8	37.8	36.1	(20)	
2%		DB	39.4	39.2	40.8	52.3	67.6	73.1	74.8	72.4	60.8	51.9	41.1	38.9	(21)	
		MCWB	34.2	33.3	33.7	40.1	50.8	55.0	57.4	57.1	48.9	43.7	35.2	33.4	(22)	
5%		DB	36.6	36.3	36.9	47.8	63.2	70.2	71.8	66.2	57.0	46.2	37.1	36.3	(23)	
		MCWB	32.2	30.8	30.7	37.9	47.5	54.1	56.6	54.9	47.1	39.6	32.6	31.3	(24)	
10%		DB	32.4	33.5	34.1	45.1	58.4	66.0	67.8	63.0	54.5	43.9	34.0	32.1	(25)	
		MCWB	28.5	28.5	28.6	36.0	45.4	52.0	55.6	52.7	46.2	37.8	30.6	28.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	36.8	37.6	44.9	53.4	60.5	61.9	61.0	52.5	47.8	38.5	37.0	(27)
MCD B			45.0	42.5	44.1	59.8	67.6	72.7	76.2	74.4	60.9	56.3	44.7	43.0	(28)	
2%		WB	35.7	33.9	34.2	40.7	51.0	57.3	59.2	58.1	50.2	43.6	35.6	34.2	(29)	
		MCD B	39.8	40.1	40.2	49.9	66.7	69.6	70.8	69.1	57.9	50.5	40.7	38.7	(30)	
5%		WB	32.7	31.1	31.2	38.4	48.4	55.1	57.4	55.6	48.5	40.1	33.1	31.5	(31)	
		MCD B	37.2	35.9	36.7	46.7	61.2	66.9	67.9	64.8	56.0	46.0	37.0	36.3	(32)	
10%		WB	29.3	28.2	28.6	36.3	46.2	53.3	55.9	53.6	46.6	38.1	30.2	28.4	(33)	
		MCD B	33.0	32.4	33.6	44.4	57.1	63.7	65.9	60.6	53.0	43.4	33.7	31.9	(34)	
(35)	Mean Daily Temperature Range	MDBR	17.1	21.4	25.3	21.0	20.5	20.2	17.8	16.6	16.5	15.6	16.0	16.7	(35)	
MCD BR			16.0	19.4	22.2	25.4	31.0	28.8	28.7	26.2	23.3	16.8	14.7	16.3	(36)	
MCWBR			12.2	15.0	15.9	15.1	16.2	14.0	14.0	13.6	13.7	10.7	11.1	12.9	(37)	
5% WB		MCD BR	16.2	18.1	18.9	22.9	27.5	24.4	23.0	21.8	19.9	15.0	14.1	15.8	(38)	
		MCWBR	12.8	14.4	13.4	13.9	14.6	12.5	12.1	11.9	11.9	9.8	10.9	12.9	(39)	
																(39)
(40)	Clear-Sky Solar Irradiance	taub	0.193	0.261	0.269	0.321	0.357	0.391	0.366	0.380	0.306	0.262	0.194	0.185	(40)	
taud		1.959	2.123	2.219	2.134	2.250	2.258	2.356	2.316	2.491	2.318	1.942	1.899	(41)		
Ebn at Noon		130	209	264	270	270	261	266	249	252	213	121	56	(42)		
Edh at Noon		9	20	27	37	37	38	33	32	22	16	9	4	(43)		
(44)	All-Sky Solar Radiation	RadAvg	61	275	849	1466	1694	1662	1430	1109	718	314	93	24	(44)	
(45)		RadStd	8	28	69	90	150	118	111	100	97	41	13	2	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+1.49	N/A	N/A	N/A	N/A	N/A	N/A	-565	N/A	N/A	N/A

Nomenclature: See separate page